Fall 1977

Problem 1 (Fa77). Let

$$A = \begin{pmatrix} 7 & 15 \\ -2 & -4 \end{pmatrix}$$

Find a real matrix B such that $B^{-1}AB$ is diagonal.

Problem 2 (Fa77, Su78).

1. Using only the axioms for a field $\mathbb{F}$, prove that a system of m homogeneous linear equations in n unknowns with $m < n$ and coefficients in $\mathbb{F}$ has a nonzero solution.

2. Use Part 1 to show that if V is a vector space over $\mathbb{F}$ which is spanned by a finite number of elements, then every maximal linearly independent subset of V has the same number of elements.

Problem 3 (Fa77). Let T be an $n \times n$ complex matrix. Show that

$$\lim_{k\to\infty} T^k = 0$$

if and only if all the eigenvalues of T have absolute value less than 1.

Problem 4 (Fa77). Let P be a linear operator on a finite dimensional vector space over a finite field. Show that if P is invertible, then $P^n = I$ for some positive integer n.

Problem 5 (Fa77).

1. In the ring $\mathbb{Z}_n$ of integers mod n, show that k is a unit if and only if k and n are relatively prime.

2. Suppose $n = pq$, where p and q are primes. Prove that the number of units in $\mathbb{Z}_n$ is $(p-1)(q-1)$.

Problem 6 (Fa77, Fa81, Fa21). Let $u : \mathbb{R}^2 \to \mathbb{R}$ be the function defined by $u(x, y) = x^3 - 3xy^2$. Show that u is harmonic and find $v : \mathbb{R}^2 \to \mathbb{R}$ such that the function $f : \mathbb{C} \to \mathbb{C}$ defined by

$$f(x + iy) = u(x, y) + iv(x, y)$$

is analytic.

Problem 7 (Fa77, Su82, Fa97, Sp01, Fa03, Fa08, Sp11, Fa12, Sp18, Sp21). Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{1 + x^{2n}}$, where n is a positive integer.

Problem 8 (Fa77). Find all solutions of the differential equation

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} - 2\frac{\mathrm{d}x}{\mathrm{d}t} + x = \sin t$$

subject to the condition $x(0) = 1$ and $x'(0) = 0$.

Problem 9 (Fa77). Let $f : [0,1] \to \mathbb{R}$ be continuously differentiable, with $f(0) = 0$. Prove that

$$\sup_{0 \leqslant x \leqslant 1} |f(x)| \leqslant \sqrt{\int_0^1 (f'(x))^2 \,\mathrm{d}x}$$

Problem 10 (Fa77, Sp80, Sp20). Let $f_n : \mathbb{R} \to \mathbb{R}$ be differentiable for each $n = 1, 2, \ldots$ with $|f_n'(x)| \leqslant 1$ for all n and x. Assume

$$\lim_{n\to\infty} f_n(x) = g(x)$$

for all x. Prove that $g : \mathbb{R} \to \mathbb{R}$ is continuous.

Problem 11 (Fa93, Fa77). Let n be an integer larger than 1. Is there a differentiable function on $[0, \infty)$ whose derivative equals its n-th power and whose value at the origin is positive?

Problem 12 (Fa77). Let $X \subset \mathbb{R}$ be a nonempty connected set of real numbers. If every element of X is rational, prove X has only one element.

Problem 13 (Fa77). Consider the following four types of transformations:

$$z \mapsto z+b \quad z \mapsto 1/z \quad z \mapsto kz \quad \text{for} \quad k \neq 0 \quad z \mapsto \frac{az+b}{cz+d} \quad \text{for} \quad ad-bc \neq$$

Here, z is a variable complex number and the other letters denote constant complex numbers. Show that each transformation takes circles to either circles or straight lines.

Problem 14 (Fa77). If a and b are complex numbers and $a \neq 0$, the set a^b consists of those complex numbers c having a logarithm of the form $b\alpha$, for some logarithm α of a. That is, $e^{b\alpha} = c$ and $e^{\alpha} = a$ for some complex number α. Describe set a^b when $a = 1$ and $b = 1/3 + i$.

Problem 15 (Fa77). Let $f : \mathbb{R}^n \to \mathbb{R}$ have continuous partial derivatives and satisfy

$$\left|\frac{\partial f}{\partial x_j}(x)\right| \leqslant K$$

for all $x = (x_1, \ldots, x_n)$, $j = 1, \ldots, n$. Prove that

$$|f(x) - f(y)| \leqslant \sqrt{n} K \|x - y\|$$

Problem 16 (Fa77). Let E and F be vector spaces. Let $S : E \to F$ be a linear transformation.

1. Prove $S(E)$ is a vector space.
2. Show S has a kernel $\{0\}$ if and only if S is injective.
3. Assume S is injective; prove $S^{-1} : S(E) \to E$ is linear.

Problem 17 (Fa77). Let G be the set of 3×3 real matrices with zeros below the diagonal and ones on the diagonal.

1. Prove G is a group under matrix multiplication.
2. Determine the center of G.

Problem 18 (Fa77). Suppose the nonzero complex number α is a zero of a polynomial of degree n with rational coefficients. Prove that $1/\alpha$ is also a zero of a polynomial of degree n with rational coefficients.

Problem 19 (Fa77). Let M be a real 3×3 matrix such that $M^3 = I$, $M \neq I$.

1. What are the eigenvalues of M?
2. Give an example of such a matrix.

Problem 20 (Fa77, Fa96, Sp19). Let $\mathbb{C}^3$ denote the set of ordered triples of complex numbers. Define a map $F : \mathbb{C}^3 \to \mathbb{C}^3$ by

$$F(u, v, w) = (u + v + w, uv + vw + wu, uvw)$$

Prove that F is onto but not one-to-one.